EXERCISE 4.5

1. Find the value, if it exists. If not, give the reason for non-existence.

   (i) $\sin^{-1}(\cos\pi)$ (ii) $\tan^{-1}\left(\sin\left(-\frac{5\pi}{2}\right)\right)$ (iii) $\sin^{-1}[\sin 5]$.

2. Find the value of the expression in terms of x, with the help of a reference triangle.

   (i) $\sin\left(\cos^{-1}(1-x)\right)$ (iii) $\cos\left(\tan^{-1}(3x-1)\right)$ (iii) $\tan\left(\sin^{-1}\left(x+\frac{1}{2}\right)\right)$.

3. Find the value of

   (i) $\sin^{-1}\left(\cos\left(\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)\right)\right)$ (ii) $\cot\left(\sin^{-1}\frac{3}{5}+\sin^{-1}\frac{4}{5}\right)$ (iii) $\tan\left(\sin^{-1}\frac{3}{5}+\cot^{-1}\frac{3}{2}\right)$.

4. Prove that

   (i) $\tan^{-1}\frac{2}{11}+\tan^{-1}\frac{7}{24}=\tan^{-1}\frac{1}{2}$ (ii) $\sin^{-1}\frac{3}{5}-\cos^{-1}\frac{12}{13}=\sin^{-1}\frac{16}{65}$.

5. Prove that $\tan^{-1}x+\tan^{-1}y+\tan^{-1}z=\tan^{-1}\left[\frac{x+y+z-xyz}{1-xy-yz-zx}\right]$.

6. If $\tan^{-1}x+\tan^{-1}y+\tan^{-1}z=\pi$, show that $x+y+z=xyz$.

7. Prove that $\tan^{-1}x+\tan^{-1}\frac{2x}{1-x^2}=\tan^{-1}\frac{3x-x^3}{1-3x^2}$, $|x|<\frac{1}{\sqrt{3}}$.

8. Simplify: $\tan^{-1}\frac{x}{y}-\tan^{-1}\frac{x-y}{x+y}$.

9. Solve:

   (i) $\sin^{-1}\frac{5}{x}+\sin^{-1}\frac{12}{x}=\frac{\pi}{2}$ (ii) $2\tan^{-1}x=\cos^{-1}\frac{1-a^2}{1+a^2}-\cos^{-1}\frac{1-b^2}{1+b^2}$, $a>0,\ b>0$.

   (iii) $2\tan^{-1}(\cos x)=\tan^{-1}(2\operatorname{cosec}x)$ (iv) $\cot^{-1}x-\cot^{-1}(x+2)=\frac{\pi}{12}$, $x>0$.

10. Find the number of solutions of the equation $\tan^{-1}(x-1)+\tan^{-1}x+\tan^{-1}(x+1)=\tan^{-1}(3x)$.

EXERCISE 4.6

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Choose the correct or the most suitable answer from the given four alternatives.

1. The value of $\sin^{-1}(\cos x)$, $0\le x\le\pi$ is

   (1) $\pi-x$ (2) $x-\frac{\pi}{2}$ (3) $\frac{\pi}{2}-x$ (4) $x-\pi$

2. If $\sin^{-1}x+\sin^{-1}y=\frac{2\pi}{3}$; then $\cos^{-1}x+\cos^{-1}y$ is equal to

   (1) $\frac{2\pi}{3}$ (2) $\frac{\pi}{3}$ (3) $\frac{\pi}{6}$ (4) π

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